



ÉCOLE POLYTECHNIQUE
FÉDÉRALE DE LAUSANNE



1

Classification of cancer types based on medical staining images

-Semester project, LTS4-

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Introduction

Goal of the project

- ▶ The patient has skin cancer
- ▶ We will have stained images of the patient's tissues and other clinical data
- ▶ We have to chose which treatment gives the best life expectancy for the patient based on these data
 - personalized treatment

Goal of the semester project

- ▶ That implies us to be able to predict the survival or (better) the lifespan of a patient after a given operation
 - machine learning: prognosis for each treatment
- ▶ Part 1: prognosis based on numerical data (done)
- ▶ Part 2: prognosis based on stained medical images (to be done)

Dataset

Visualization of the dataset

bcr_patient_uuid	bcr_patient_barcode	form_completion_date	prospective_collection	retrospective_collection	birth_days_to	gender
bcr_patient_uuid	bcr_patient_barcode	form_completion_date	tissue_prospective_collection_indication	tissue_retrospective_collection_indication	days_to_birth	gender
CDE_ID:	CDE_ID:2673794	CDE_ID:	CDE_ID:3088492	CDE_ID:3088528	CDE_ID:3008233	CDE_ID:2200604
5564E6A7-2195-4B0D-994E-B0617B58E889	TCGA-3N-A9WB	2014-5-29	YES	NO	-26176	MALE
551E071A-C290-4B48-9000-F64C2A44DFB7	TCGA-3N-A9WC	2014-5-29	YES	NO	-30286	MALE
A29A20E3-5C2C-4F37-B93E-AE9EBC46EC53	TCGA-3N-A9WD	2014-5-29	YES	NO	-30287	MALE
3DD5A206-D7F3-42F1-B9CC-4B31C76D495D	TCGA-BF-A1PU	2013-4-25	YES	NO	-30288	FEMALE
EFF78AF6-0F68-49B9-866B-0D511606F2B1	TCGA-BF-A1PV	2013-4-25	YES	NO	-30289	FEMALE
197AC33E-D5DF-40DE-92F2-6EF7FE2AD6D1	TCGA-BF-A1PX	2013-4-25	YES	NO	-30290	MALE
455F982C-A067-46AC-BF89-3E535D0FFCA0	TCGA-BF-A1PZ	2013-4-25	YES	NO	-30291	FEMALE
633DDC04-C424-42C8-8E54-673968B130A7	TCGA-BF-A1Q0	2013-4-25	YES	NO	-30292	MALE
62393DAB-AB41-4785-9A57-AA4C25D223E9	TCGA-BF-A3DJ	2013-4-25	YES	NO	-30293	FEMALE
E9F0F030-D9AA-47C8-9F4D-A9CE58888C41	TCGA-BF-A3DL	2013-4-25	YES	NO	-30294	FEMALE
1CA2EB9F-9496-450F-83FE-C7A6CB1A075B	TCGA-BF-A3DM	2013-4-25	YES	NO	-30295	MALE
D7E86C99-A716-48B8-B067-E73D68B14DEE	TCGA-BF-A3DN	2013-4-25	YES	NO	-29801	FEMALE
8352CBB0-10E7-40BD-96F0-1E982AB4C07B	TCGA-BF-A5EO	2014-11-13	YES	NO	-23940	MALE

Visualization of the dataset

- ▶ 470 patients
- ▶ 62 columns

icd_10	icd_o_3_histolog	icd_o_3_site	informed consent verification	patient_id	project_code	stage_other	tissue_source_site
CDE_ID:3226287	CDE_ID:3226275	CDE_ID:3226281	Quartier et Cie: The third edition of the International Classification of Diseases for Oncology, published in 2000, used principally in tumor and cancer registries for coding the site (topography) and the histology (morphology) of neoplasms. The description of an anatomical region or of a body part. Named locations of, or within, the body. A system of numbered categories for representation of data.	CDE_ID:	CDE_ID:	CDE_ID:2007104	CDE_ID:
C44.5	8720/3	C44.5	YES	A9WB	[Not Available]	[Not Available]	3N
C77.3	8720/3	C77.3	YES	A9WC	[Not Available]	[Not Available]	3N
C77.0	8720/3	C77.0	YES	A9WD	[Not Available]	[Not Available]	3N
C44.9	8720/3	C44.9	YES	A1PU	[Not Available]	[Not Available]	BF
C44.9	8720/3	C44.9	YES	A1PV	[Not Available]	[Not Available]	BF
C44.9	8720/3	C44.9	YES	A1PX	[Not Available]	[Not Available]	BF
C44.9	8720/3	C44.9	YES	A1PZ	[Not Available]	[Not Available]	BF
C44.9	8720/3	C44.9	YES	A1Q0	[Not Available]	[Not Available]	BF
C44.9	8720/3	C44.9	YES	A3DJ	[Not Available]	[Not Available]	BF
C44.9	8720/3	C44.9	YES	A3DL	[Not Available]	[Not Available]	BF
C44.9	8720/3	C44.9	YES	A3DM	[Not Available]	[Not Available]	BF
C44.9	8720/3	C44.9	YES	A3DN	[Not Available]	[Not Available]	BF
C44.5	8720/3	C44.5	YES	A5EO	[Not Available]	[Not Available]	BF

Visualization of the dataset

- ▶ Three essential columns:
 - ▶ Vital status
 - ▶ Days to last followup: time interval from the date of last followup to the date of initial pathologic diagnosis
 - ▶ Patients still alive
 - ▶ Days to death: time interval from a person's date of death to the date of initial pathologic diagnosis
 - ▶ Patients already dead

vital_status	last_contact_days_to	death_days_to
vital_status	days_to_last_followup	days_to_death
CDE_ID:5	CDE_ID:3008273	CDE_ID:3165475
Dead	[Not Available]	518
Alive	1856	[Not Applicable]
Dead	[Not Available]	395
Alive	387	[Not Applicable]
Alive	14	[Not Applicable]
Dead	[Not Available]	282
Alive	12	[Not Applicable]
Alive	17	[Not Applicable]
Alive	464	[Not Applicable]
Alive	28	[Not Applicable]
Alive	14	[Not Applicable]
Alive	32	[Not Applicable]
Alive	338	[Not Applicable]

Pre-processing

Pre-processing

→ Discard meaningless columns

```
vital_status',  
'days_to_last_followup',  
'days_to_death',  
'tissue_prospective_collection_indicator',  
'tissue_retrospective_collection_indicator',  
'days_to_birth',  
'gender',  
'height',  
'weight',  
'race',  
'ethnicity',  
'other_dx',  
'history_of_neoadjuvant_treatment',  
'person_neoplasm_cancer_status',  
'primary_neoplasm_melanoma_dx',  
'primary_tumor_multiple_present_ind',  
'primary_melanoma_at_diagnosis_count',  
'breslow_depth_value',  
'melanoma_clark_level_value',  
'melanoma_ulceration_indicator',  
'malignant_neoplasm_mitotic_count_rate',  
'age_at_initial_pathologic_diagnosis',  
'pathologic_T',  
'pathologic_N',  
'pathologic_M',  
'lactate_dehydrogenase_result',  
'pathologic_stage',  
'days_to_submitted_specimen_dx',  
'tumor_tissue_site',  
'distant_metastasis_anatomic_site',  
'melanoma_origin_skin_anatomic_site',  
'radiation_therapy',  
'postoperative_rx_tx',  
'new_tumor_event_after_initial_treatment',  
'new_tumor_dx_prior_submitted_specimen_dx',  
'primary_anatomic_site_count',  
'clinical_M',  
'clinical_N',  
'clinical_T',  
'clinical_stage',  
'icd_10',  
'icd_o_3_histology',  
'icd_o_3_site',  
'stage_other',
```

Pre-processing

- Discard patients of which we don't know the vital status or the duration of study

List of the patients of which the survival time is missing or incorrect:

[55, 56, 71, 95, 113, 143, 147, 148, 151, 156, 164, 165, 166, 167, 168, 169, 176, 177, 181, 186, 187, 188, 191, 192, 309, 333, 426, 427, 463]

- Transform the two columns « days to last followup » and « days to death » into one column (easier to handle)

vital_status	last_contact_days	to_death_days	to
vital_status	days_to_last_followup	days_to_death	
CDE_ID:5	CDE_ID:3008273	CDE_ID:3165475	
Dead	[Not Available]	518	
Alive	1856	[Not Applicable]	
Dead	[Not Available]	395	
Alive	387	[Not Applicable]	
Alive	14	[Not Applicable]	
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Alive	12	[Not Applicable]	
Alive	17	[Not Applicable]	
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Alive	28	[Not Applicable]	
Alive	14	[Not Applicable]	
Alive	32	[Not Applicable]	
Alive	338	[Not Applicable]	

Pre-processing

- ▶ Numerical features
- ▶ Binary features
- ▶ Multicategorical features → several binary features

gender	weight_kg_at_diag	race
gender	weight	race
CDE_ID:2200604	CDE_ID:651	CDE_ID:2192199
MALE	113	WHITE
FEMALE	57	BLACK OR AFRICAN AMERICAN
FEMALE	83	WHITE
MALE	73	WHITE
FEMALE	81	WHITE
MALE	80	WHITE
FEMALE	89	WHITE
MALE	63	WHITE
FEMALE	53	WHITE
FEMALE	43	ASIAN
MALE	50	ASIAN

Pre-processing

- Fill in the empty cases of the Excel table
 - Is the void meaningful?
 - If not, fill with the mean of the column

malignant_neoplasm_mitotic_count_r
CDE_ID:3119292
2
4
4
[Not Available]
[Not Available]
10
[Not Available]
[Not Available]
[Not Available]
[Not Available]
[Not Available]
[Not Available]

ethnicity
CDE_ID:2192217
NOT HISPANIC OR LATIN
NOT HISPANIC OR LATIN
NOT HISPANIC OR LATIN
[Unknown]
[Unknown]
NOT HISPANIC OR LATIN
NOT HISPANIC OR LATIN
NOT HISPANIC OR LATIN
[Unknown]
NOT HISPANIC OR LATIN
NOT HISPANIC OR LATIN
HISPANIC OR LATINO

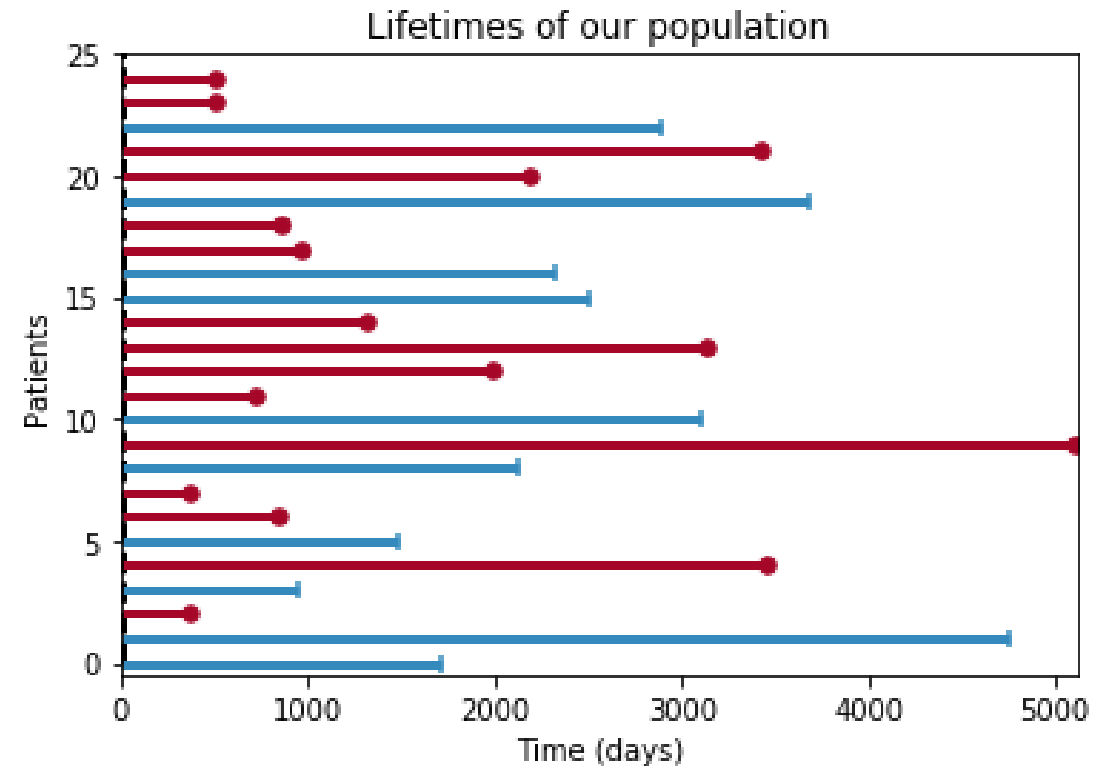
Pre-processing

- ▶ Before pre-processing:
 - ▶ 470 patients
 - ▶ 62 columns
 - ▶ 3 columns: « vital status », « days to last followup », « days to death »
- ▶ After pre-processing:
 - ▶ 441 patients
 - ▶ 265 features
 - ▶ 2 essential features: « vital status » and « duration »

Survival estimation

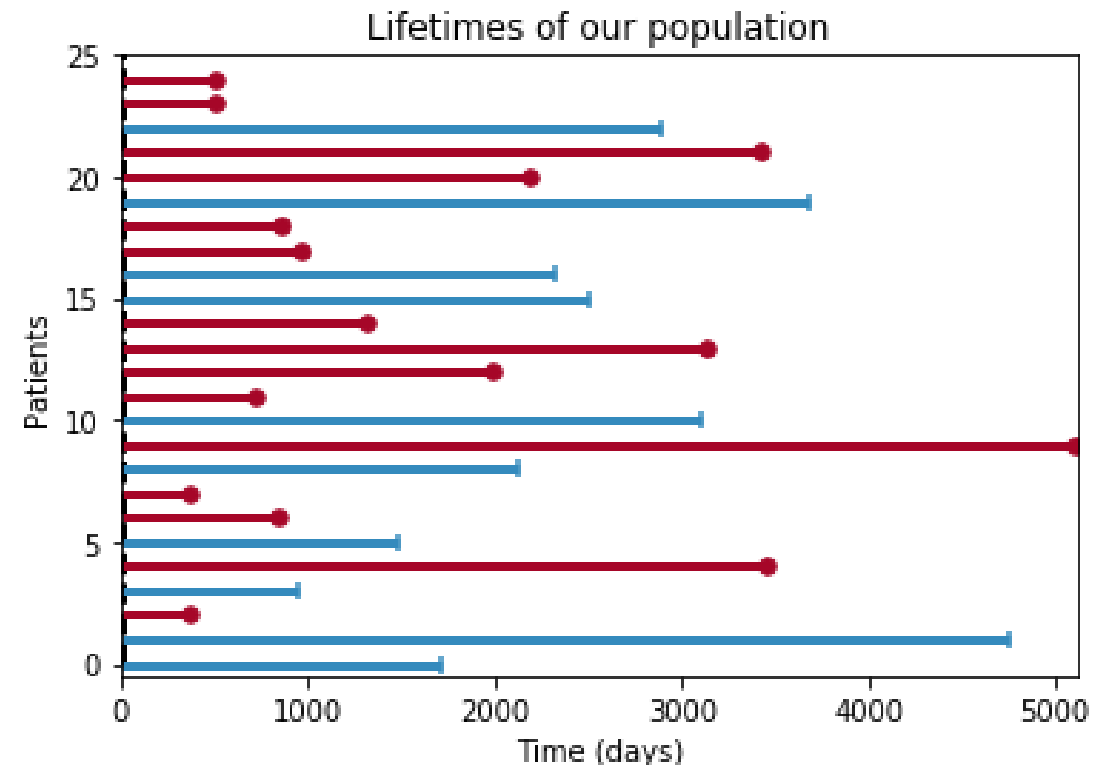
Why survival analysis?

- ▶ We want to make inferences about durations but not all the death events have occurred yet (65.1% of the population still alive)



Why survival analysis?

- ▶ Can't use classical machine learning algorithms
- ▶ If we ignore the right-censored data: underestimation!
- ▶ Survival analysis^[1]



^[1] Cox, D., Reid, N. (1984). *Analysis of Survival Data*. New York: Routledge.

Survival analysis

- ▶
- ▶ Two fundamental mathematical objects:
 - ▶ survival function $S(t)$
 - ▶ hazard function $\lambda(t)$

Survival analysis: survival function

- ▶ Let T be the random variable of the lifespan of an individual, and t be a random lifetime taken from the population under study.
- ▶ The survival function $S(t)$ of the individual is defined as the probability that the death event has not occurred yet at time t or, equivalently, as the probability of surviving past time t :

$$S(t) = \Pr(T > t)$$

Survival analysis: hazard function

- The hazard function $\lambda(t)$ at time t is the probability of the death event occurring at time t , given that the death event has not occurred until time t :

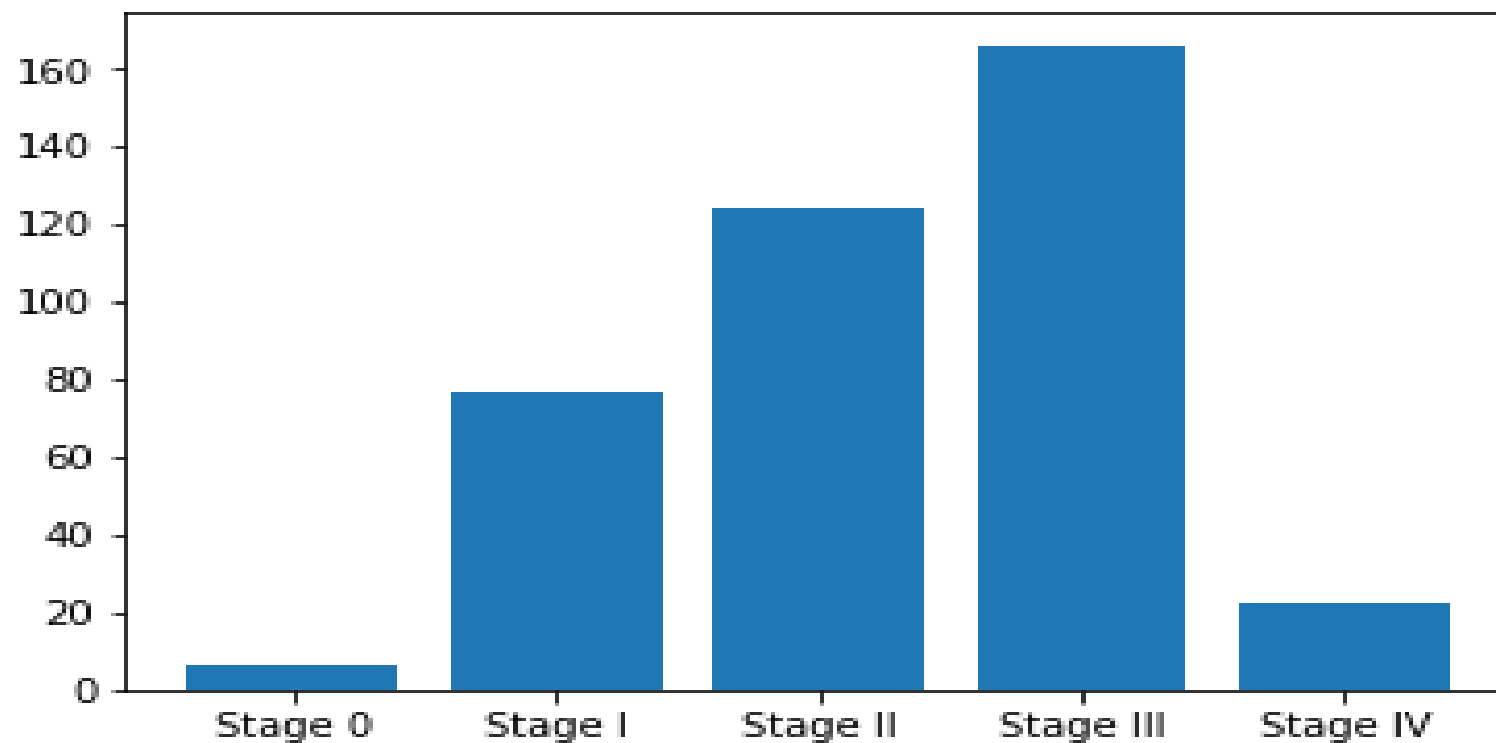
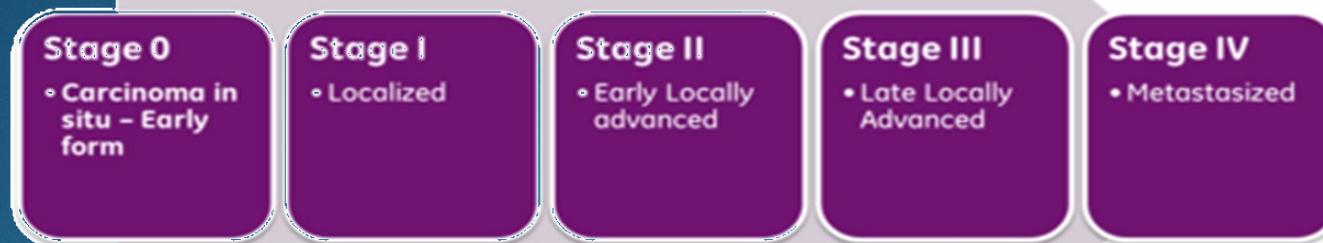
$$\lambda(t) = \lim_{\delta t \rightarrow 0} \frac{\Pr(t \leq T \leq t + \delta t | T > t)}{\delta t}$$

- Note that we can switch from a function to another quite easily:

$$\lambda(t) = -\frac{S'(t)}{S(t)}; \quad S(t) = \exp\left(-\int_0^t \lambda(z) dz\right)$$

Survival analysis: simple estimators

- ▶ Hypothesis on our dataset:
 - ▶ The survival probabilities are the same for subjects recruited early and late in the study
 - ▶ The pattern of censoring is independent of the survival times
 - ▶ The death events happen at the specified times



Survival analysis: simple estimators

- ▶
- ▶ We would like to estimate the survival function
→ Kaplan-Meier estimator^[1]:

$$\hat{S}(t) = \prod_{t_i < t} \left(\frac{n_i - d_i}{n_i} \right)$$

where d_i is the number of death events at time t_i

n_i the number of subjects at risk of death just prior to time t_i .

vital_status	last_contact_days_to	death_days_to
vital_status	days_to_last_followup	days_to_death
CDE_ID:5	CDE_ID:3008273	CDE_ID:3165475
Dead	[Not Available]	518
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Alive	28	[Not Applicable]
Alive	14	[Not Applicable]
Alive	32	[Not Applicable]
Alive	338	[Not Applicable]

^[1] Efron, B. (1988). Logistic regression, survival analysis, and the Kaplan-Meier curve. *Journal of the American statistical Association*, 83(402), 414-425.

Survival analysis: simple estimators

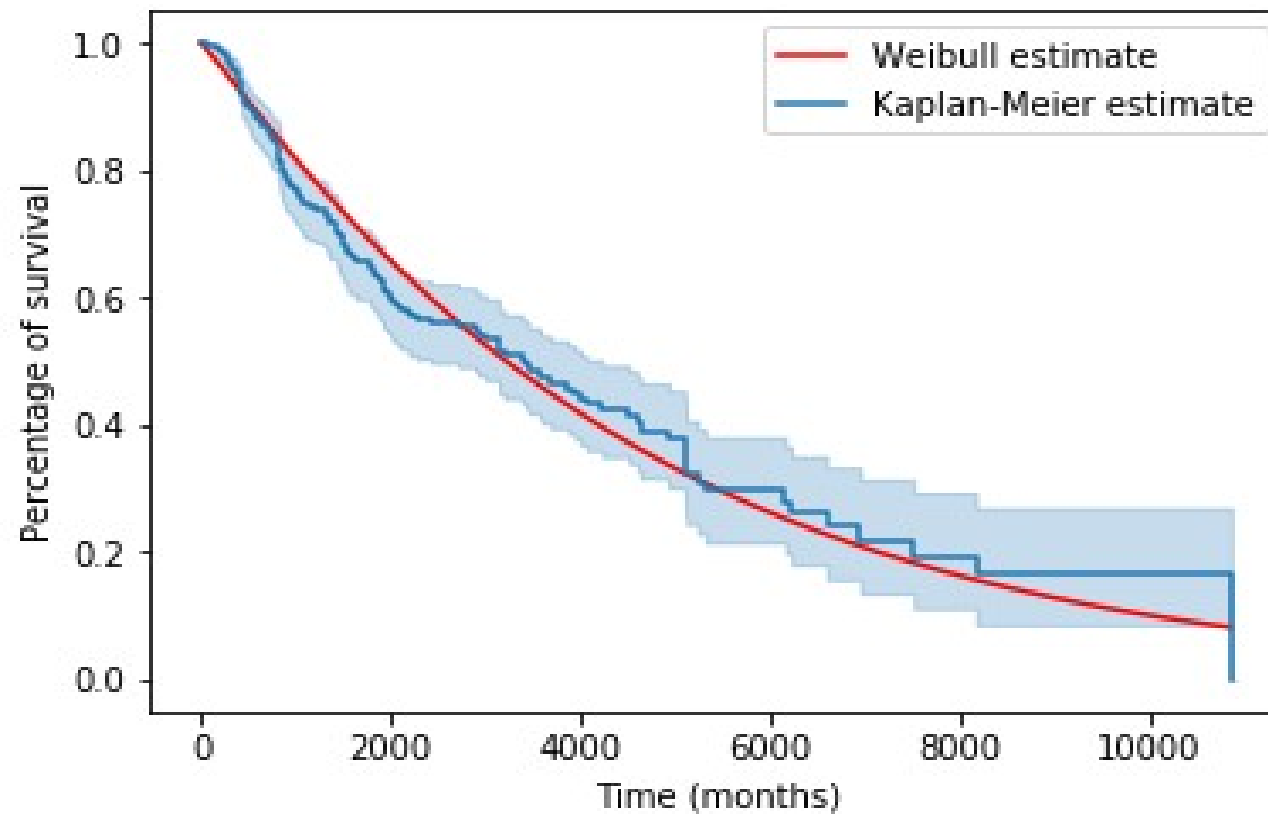
- ▶
- ▶ Another very popular model for survival data: Weibull model^[1]
 - Parametrical, with two parameters λ and ρ :

$$\hat{S}(t) = e^{-(\lambda \cdot t)^\rho}$$

vital_status	last_contact_days_to	death_days_to
vital_status	days_to_last_followup	days_to_death
CDE_ID:5	CDE_ID:3008273	CDE_ID:3165475
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Alive	14	[Not Applicable]
Alive	32	[Not Applicable]
Alive	338	[Not Applicable]

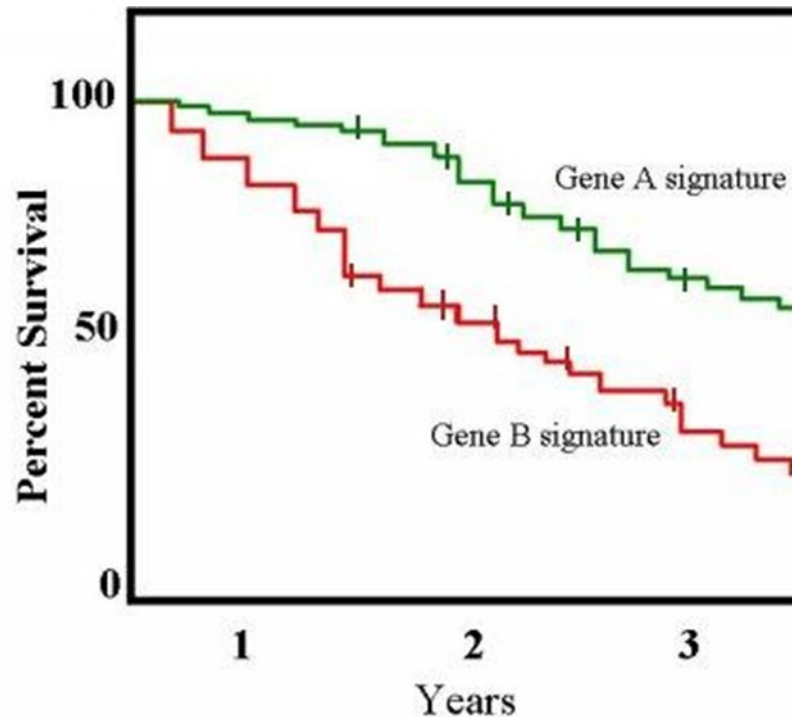
^[1] Carroll, K. J. (2003). On the use and utility of the Weibull model in the analysis of survival data. *Controlled clinical trials*, 24(6), 682-701.

Comparison: Kaplan-Meier vs Weibull



Application to clustering

- ▶ We would like to discriminate groups of patients which have very different survival functions and keep interpretability of the separation criteria
- ▶ Example:



Application to clustering

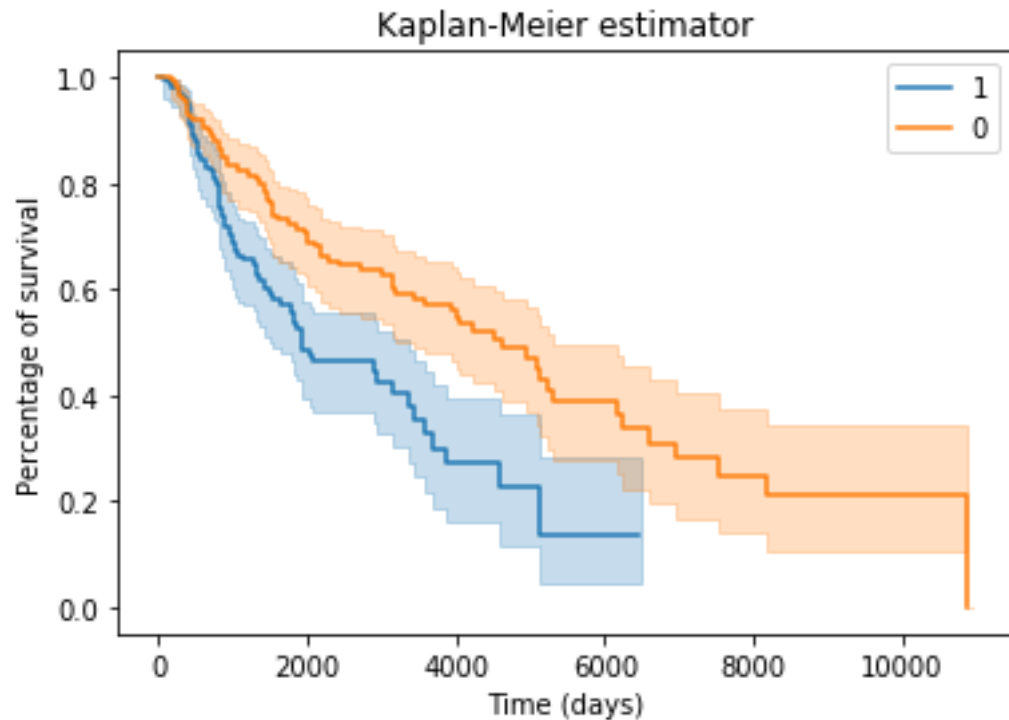
→ Goal: cluster the data into several groups with different Kaplan-Meier curves

→ K-Means clustering on the dataset (without the vital status and the duration columns)

```
vital_status',  
'days_to_last_followup',  
'days_to_death',  
'tissue_prospective_collection_indicator',  
'tissue_retrospective_collection_indicator',  
'days_to_birth',  
'gender',  
'height',  
'weight',  
'race',  
'ethnicity',  
'other_dx',  
'history_of_neoadjuvant_treatment',  
'person_neoplasm_cancer_status',  
'primary_neoplasm_melanoma_dx',  
'primary_tumor_multiple_present_ind',  
'primary_melanoma_at_diagnosis_count',  
'breslow_depth_value',  
'melanoma_clark_level_value',  
'melanoma_ulceration_indicator',  
'malignant_neoplasm_mitotic_count_rate',  
'age_at_initial_pathologic_diagnosis',  
'pathologic_T',  
'pathologic_N',  
'pathologic_M',  
'lactate_dehydrogenase_result',  
'pathologic_stage',  
'days_to_submitted_specimen_dx',  
'tumor_tissue_site',  
'distant_metastasis_anatomic_site',  
'melanoma_origin_skin_anatomic_site',  
'radiation_therapy',  
'postoperative_rx_tx',  
'new_tumor_event_after_initial_treatment',  
'new_tumor_dx_prior_submitted_specimen_dx',  
'primary_anatomic_site_count',  
'clinical_M',  
'clinical_N',  
'clinical_T',  
'clinical_stage',  
'icd_10',  
'icd_o_3_histology',  
'icd_o_3_site',  
'stage_other',
```

Application to clustering

- Goal: cluster the data into several groups with different Kaplan-Meier curves
- K-Means clustering on the dataset (without the vital status and the duration columns)



Survival regression

Survival regression

- ▶ We now include directly the additional data aside from the duration to survival analysis
 - survival regression^[1]
- ▶ Two popular competing yet complementary techniques in survival regression:
 - ▶ Cox's Proportional Hazard model
 - ▶ Aalen's Additive Model

^[1] Perme, M. P., Henderson, R., & Stare, J. (2008). An approach to estimation in relative survival regression. *Biostatistics*, 10(1), 136-146.

Survival regression: Cox's model

- ▶ Cox's Proportional Hazard model^[1]:

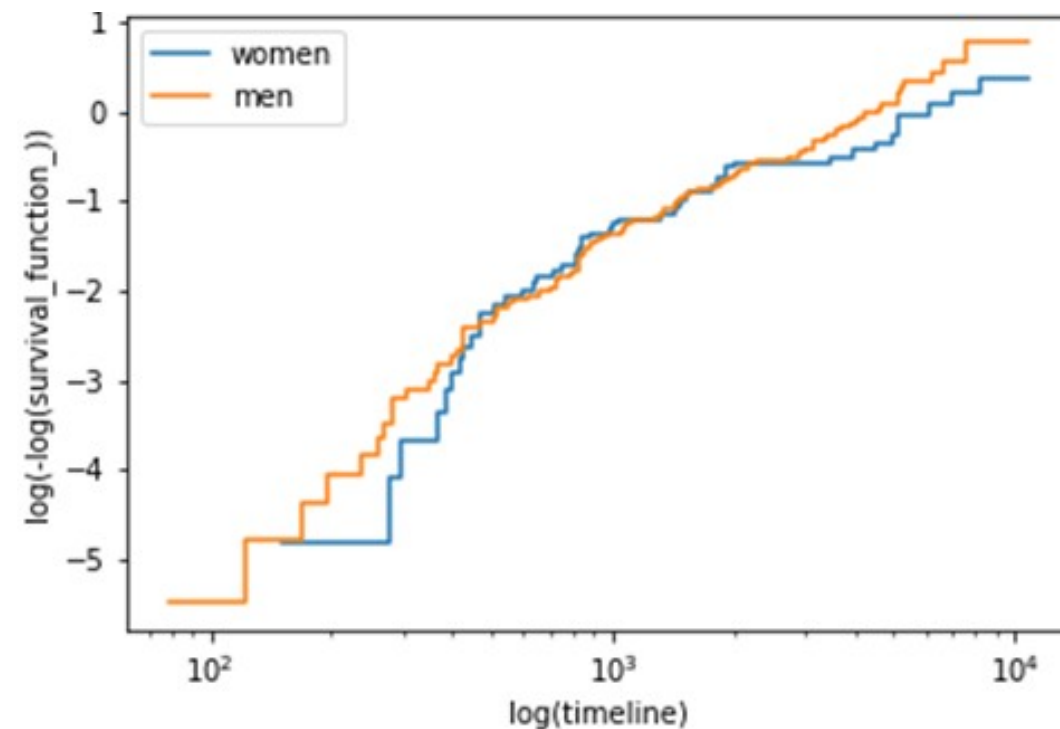
$$\lambda(t|\mathbf{x}) = b_0(t) \cdot \exp\left(\sum_{i=1}^d b_i x_i\right)$$

- ▶ Proportional hazard assumption has to be verified:
 - ▶ The only time component is in the baseline hazard, $b_0(t)$
 - ▶ The term in exp is only a scalar factor that only decreases or increases the baseline hazard

^[1] Cox, D., Reid, N. (1984). *Analysis of Survival Data*. New York: Routledge.

Survival regression: Cox's model

- ▶ Example: verification of the proportional hazard assumption on the variable « gender »:
- ▶ We tried to discard the variables that do not satisfy the assumption (=stratification), but it didn't yield good results
→ Aalen's model



Survival regression: Aalen's model

- ▶ Aalen's Additive model^[1]:

$$\lambda(t|\mathbf{x}) = b_0(t) + b_1(t).x_1 + \dots + b_d(t).x_d$$

- ▶ No need for any assumption
- ▶ L2 penalizer term

^[1] Martinussen, T., Vansteelandt, S., Gerster, M., & Hjelmberg, J. V. B. (2011). Estimation of direct effects for survival data by using the Aalen additive hazards model. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 73(5), 773-788

Survival regression: Aalen's model

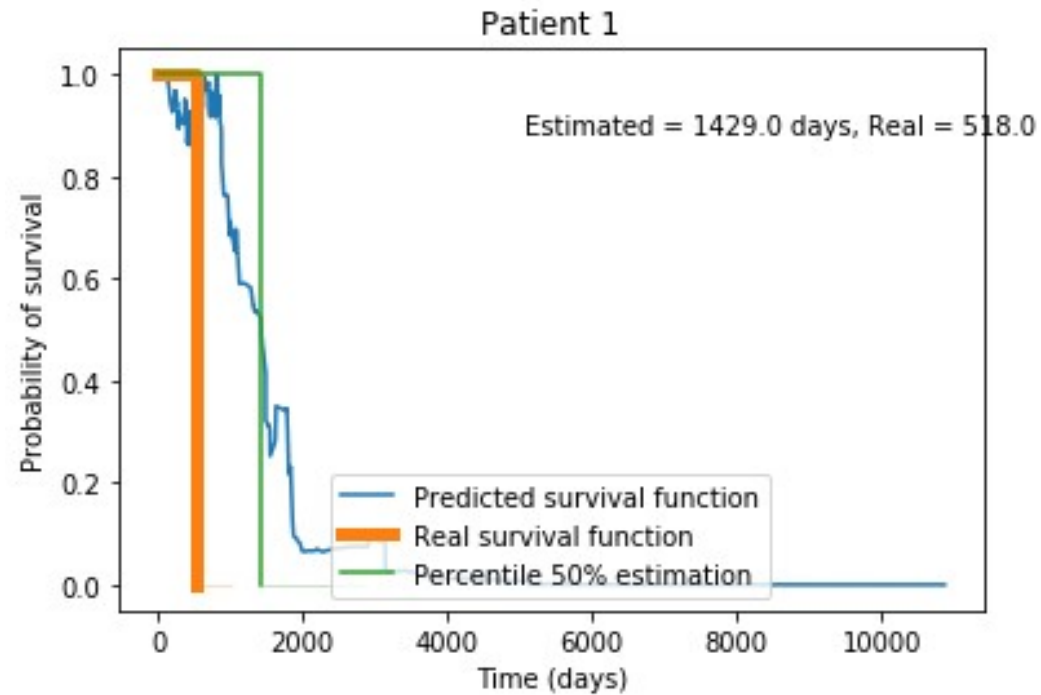
- ▶ ! Might give negative hazards^[1] !

^[1] Sasieni, P. D., & McKeague, I. W. (1992). Information bounds for the additive and multiplicative intensity models. In *Survival analysis: state of the art* (pp. 249-265). Springer, Dordrecht.

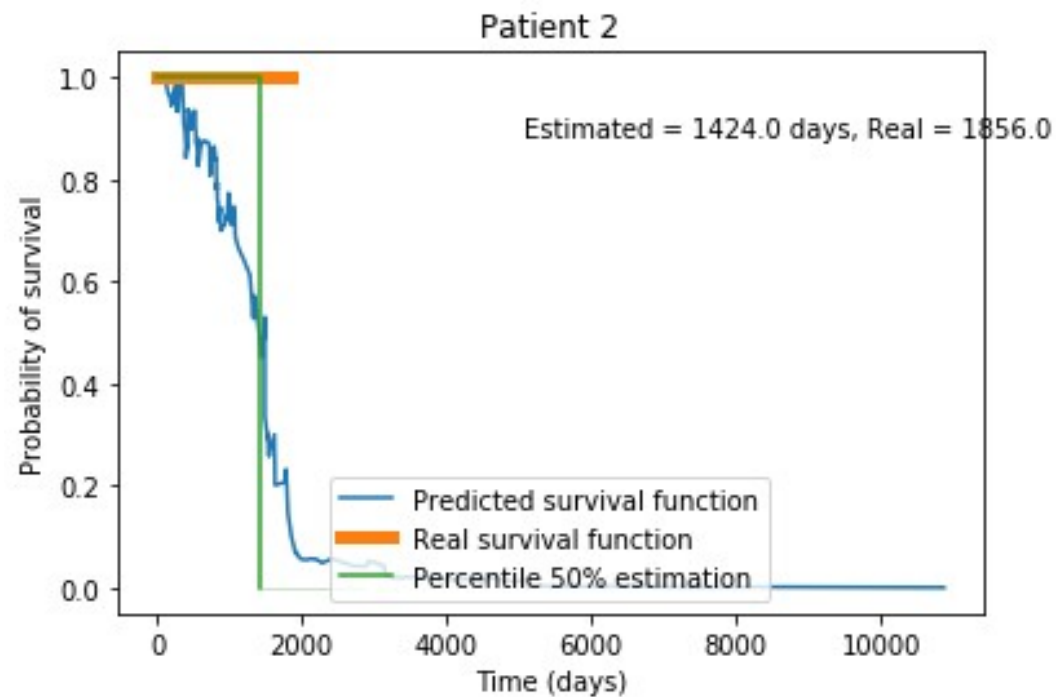
Application to prediction

- ▶ LOOCV (Leave-Out One Cross-Validation) to train and test the Aalen's model
 - ▶ Examples: computed the predicted survival functions for 4 patients

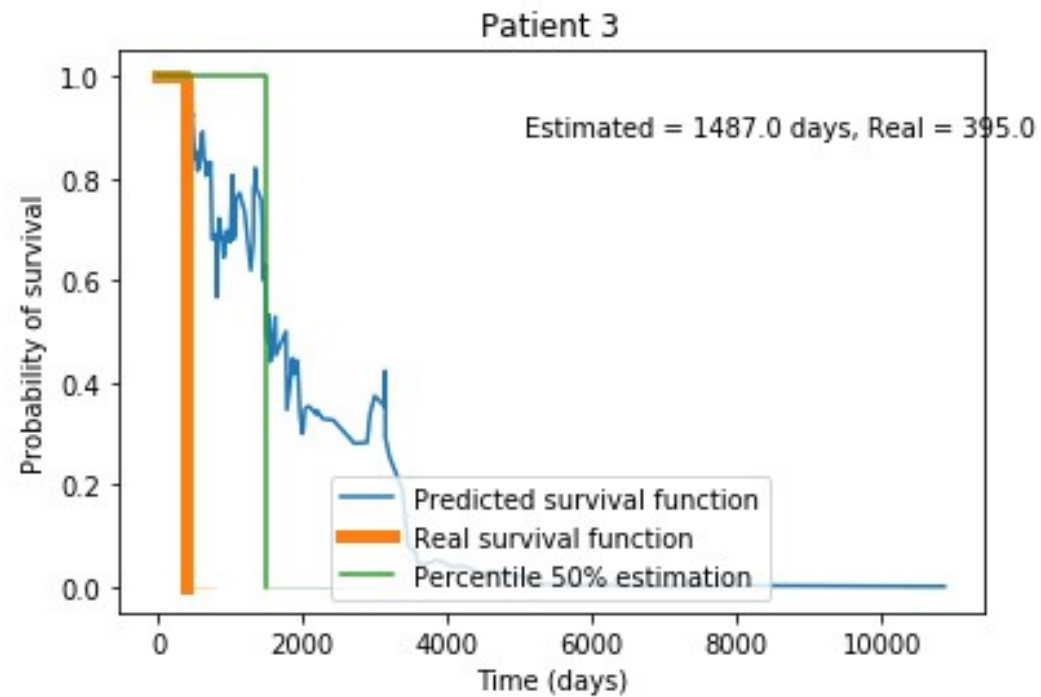
Application to prediction



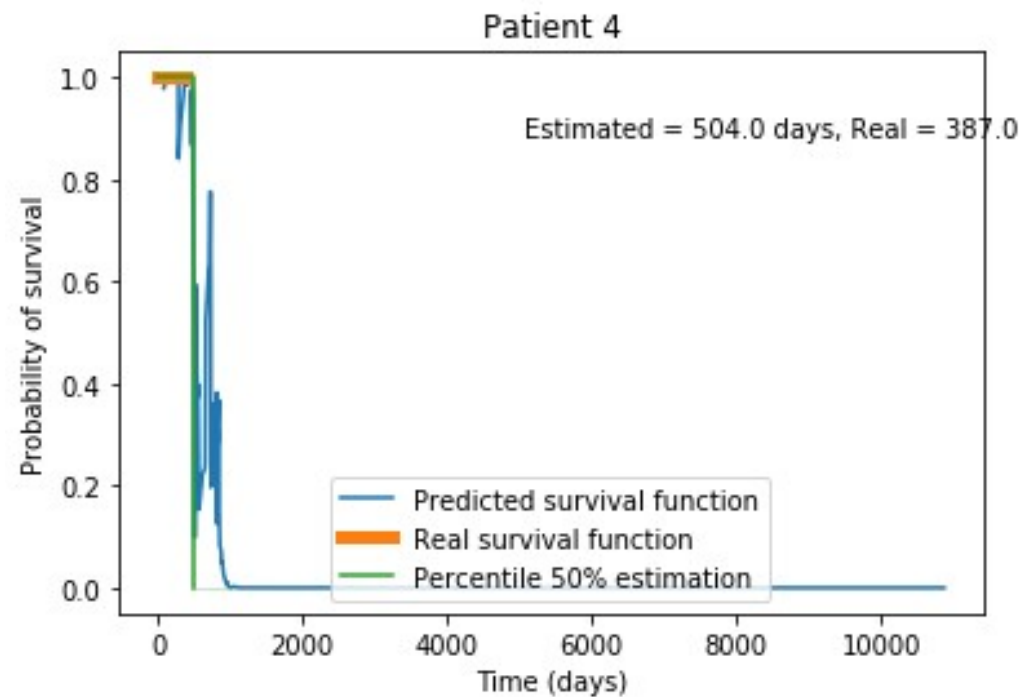
Application to prediction



Application to prediction



Application to prediction

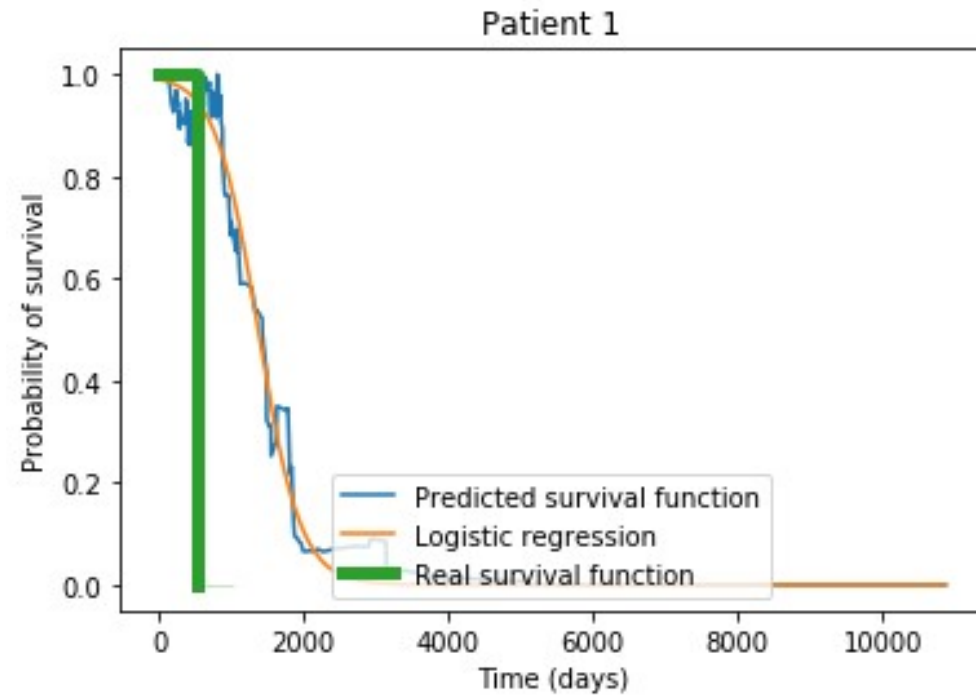


Application to prediction

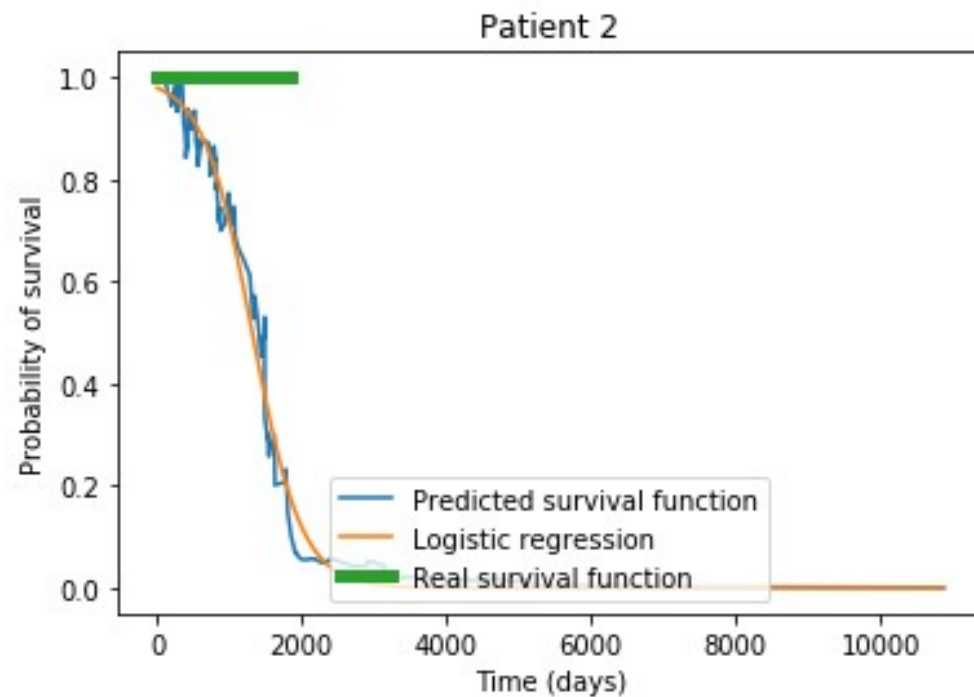
- ▶ Try to fit a logistic function to the predicted survival functions

$$y = \frac{1}{1 + \exp(-k\tilde{\mathbf{r}}\mathbf{x} - x_0\tilde{\mathbf{r}})}$$

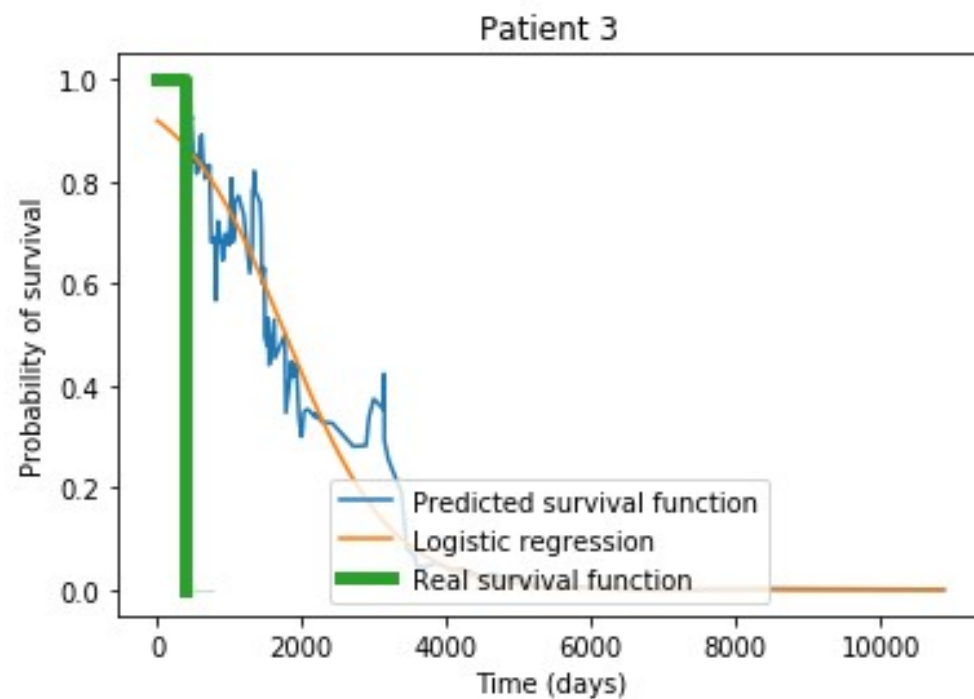
Application to prediction



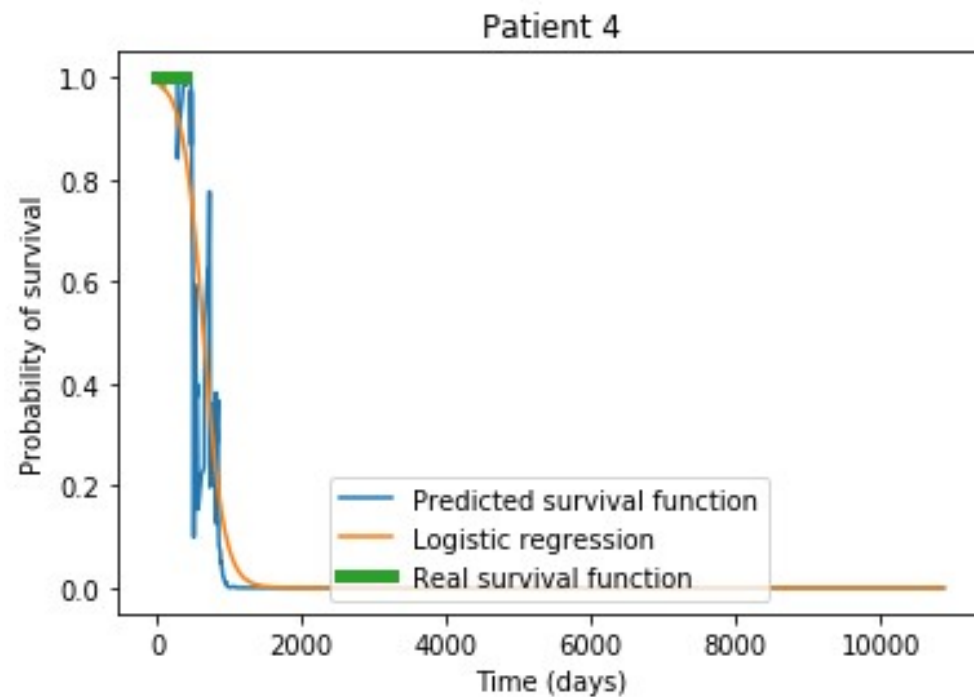
Application to prediction



Application to prediction



Application to prediction

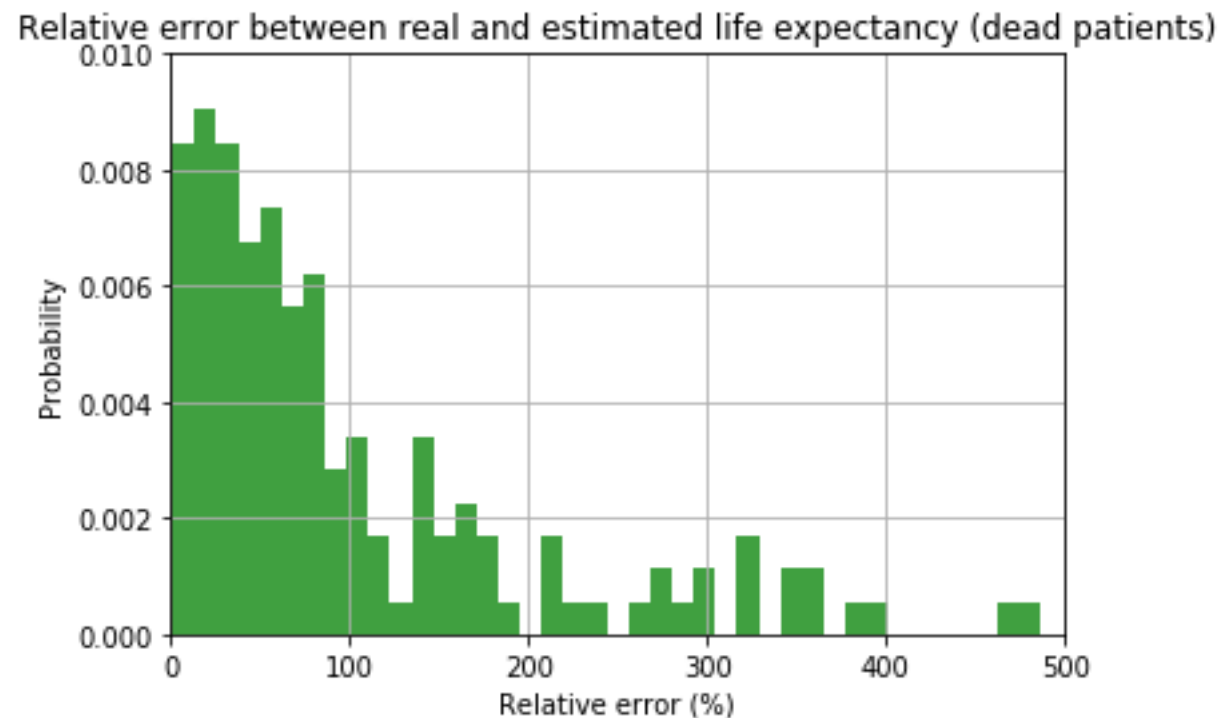


Application to prediction

- ▶ Not a very good correlation with the real survival functions
- ▶ Different criteria depending on the vital status of the patients:
 - ▶ If already dead: estimate $\sim$ real ?
 - ▶ If still alive: estimate $>$ real ?

Application to prediction

- ▶ Results:
 - ▶ If already dead:
average relative error 119%
 - ▶ If still alive:
test score 36.7%



Conclusion and perspectives

Conclusion

- ▶ Trying to predict directly the survival function of a patient with survival regression didn't yield good results
- ▶ Best way: cluster the data into groups of very different life expectancies and small confidence intervals

Perspectives

- ▶ Ask for information on more patients → bigger dataset
- ▶ Extract meaningful features from medical stained images of the patients' tissues